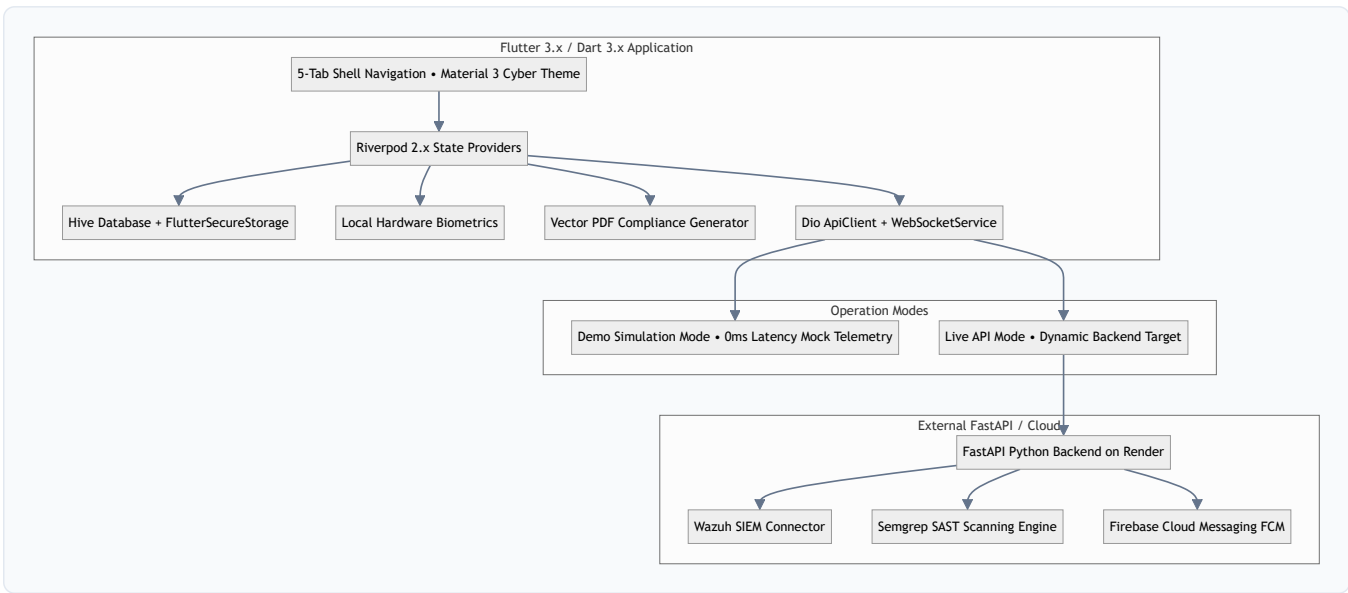


SecurePulse Mobile — Master Documentation Audit & Implementation Status



Audit Date	Repository Target	Codebase Version	Test Suite Status	Working Tree
2026-09-01	NATTOMR/securepulse-mobile	1.0.0+1	● 15/15 Passing (100%)	● Clean (main)

1. System Architecture Overview



2. Feature Implementation Matrix

Every feature below has been directly verified against the active source code, configuration files, network clients, UI screens, and test suites.

Feature Area	Status	Source Verification Path	Actual State & Verification Notes
Dual Operation Mode (Demo vs Live)	[IMPLEMENTED]	lib/core/config/app_config.dart , lib/features/settings/presentation/settings_screen.dart	Dynamic toggle in UI backed by SharedPreferences . Controls whether repositories return local mock telemetry or query FastAPI endpoints.
State Management Architecture	[IMPLEMENTED]	lib/providers/app_providers.dart	Riverpod 2.6.1 providers managing auth, dashboard telemetry, alerts, repositories, scans, and settings.
Declarative Shell Navigation	[IMPLEMENTED]	lib/core/router/app_router.dart	GoRouter shell route hosting 5 primary bottom tabs (Dashboard , Repositories , AI Assistant , SOC Alerts , Settings) + 6 modal subroutes.
Executive Security Dashboard	[IMPLEMENTED]	lib/features/dashboard/presentation/dashboard_screen.dart	Displays composite posture gauge (88-94%), vulnerability breakdown donut chart (flutter_chart), recent security events, and service status monitors.
Local Hardware Biometrics	[IMPLEMENTED]	lib/core/services/biometric_service.dart , android/app/src/main/AndroidManifest.xml	Integrates local_auth with USE_BIOMETRIC and USE_FINGERPRINT permissions to lock/unlock app sessions.
Local Encrypted Storage	[IMPLEMENTED]	lib/core/storage/secure_storage_service.dart , lib/core/storage/hive_storage_service.dart	AES-256 / RSA hardware keychain token storage (flutter_secure_storage) and key-value database caching (hive_flutter).
REST API Client & Interceptors	[IMPLEMENTED]	lib/core/network/api_client.dart , lib/core/error/api_exception.dart	Dio client with auto JWT bearer injection, timeout configurations (12s), custom base URL mutation, and structured error categorization.
Real-Time WebSocket Incident Stream	[IMPLEMENTED]	lib/core/network/websocket_service.dart	Subscribes to /ws/alerts , converts http→ws and https→wss , handles connection state broadcast, and falls back to HTTP polling if blocked.
Codebase & SAST Scan Triggering	[IMPLEMENTED]	lib/features/repositories/data/repository_repository.dart , lib/features/scans/scans_screen.dart	Monitored repository list, health scores (A-F), finding counters, and on-demand scan triggers (/v1/repositories/{id}/scan).
AI Cybersecurity Copilot	[IMPLEMENTED]	lib/features/ai/data/ai_repository.dart , lib/features/ai/presentation/ai_assistant_screen.dart	Interactive chat UI with markdown code syntax. Generates CVE-2024-3094, SQLi, and secret exposure remediation playbooks locally in Demo mode, or forwards to /v1/ai/chat in Live mode.
SOC SIEM Alert Triage	[IMPLEMENTED]	lib/features/alerts/data/alerts_repository.dart , lib/features/alerts/presentation/alert_detail_screen.dart	SIEM incident list filtered by severity (Critical, High, Medium, Low), detailed triage with origin IP and raw payload, and status updates (/v1/soc/alerts/{id}/status).
Vector PDF Compliance	[IMPLEMENTED]	lib/core/services/report_pdf_service.dart , lib/features/reports/reports_screen.dart	Pure vector PDF engine generating SOC 2 Type II, ISO 27001, PCI-DSS

Feature Area	Status	Source Verification Path	Actual State & Verification Notes
Generator			v4.0, and HIPAA audit PDFs with SHA256 audit fingerprint stamps and native share sheet export.
Diagnostic & Environment Switcher	[IMPLEMENTED]	lib/features/settings/presentation/settings_screen.dart	Allows switching between Render Cloud (https://secureguard-backend-7eqm.onrender.com), Android Emulator (http://10.0.2.2:8000), Localhost (http://127.0.0.1:8000), or custom endpoints with live ping validation.
Dual Theme System (Dark / Light)	[IMPLEMENTED]	lib/core/theme/app_theme.dart	Cyber Dark Obsidian & Clean Light themes with Google Fonts Inter and custom status color tokens.
Automated Test Coverage	[IMPLEMENTED]	test/api_integration_test.dart , test/widget_test.dart	15/15 automated tests verifying API contracts, error classifications, repository mode isolation, WebSocket URL transformations, and widget pump.
Web Deployment Container	[IMPLEMENTED]	Dockerfile	Multi-stage / Nginx Alpine container serving built Flutter web bundle on port 80.
Firebase Cloud Messaging (FCM)	[PARTIALLY IMPLEMENTED]	lib/core/services/notification_service.dart	Handlers for permissions, topic subscriptions (<code>soc_critical</code> , <code>wazuh_alerts</code>), and foreground/background streams are fully implemented; requires user to place valid <code>google-services.json</code> in <code>android/app/</code> for cloud push delivery. Falls back to local in-app stream.
GitHub OAuth Browser Redirect	[PARTIALLY IMPLEMENTED]	lib/features/auth/presentation/login_screen.dart	UI trigger exists; currently completes auth using analyst demo session rather than external browser deep-link redirect.
Direct SOAR Firewall Execution	[PARTIALLY IMPLEMENTED]	lib/features/alerts/presentation/alert_detail_screen.dart	UI action buttons ("Quarantine IP", "Acknowledge") perform alert status updates against backend; edge firewall rule orchestration is delegated to backend webhook listeners.
Two-Way SIEM Ticketing (Jira / ServiceNow)	[PLANNED]	docs/ROADMAP.md	Phase 4 roadmap feature for bidirectional sync with external IT service management platforms.
Offline Action & Mutation Queue	[PLANNED]	docs/ROADMAP.md	Phase 4 roadmap feature for queueing mitigation actions while offline and flushing upon reconnect.
In-App Custom Semgrep Rule Editor	[PLANNED]	docs/ROADMAP.md	Phase 4 roadmap feature for live YAML authoring and syntax testing within the mobile app.

Feature Area	Status	Source Verification Path	Actual State & Verification Notes
Zero-Trust Client mTLS Certificates	[NOT IMPLEMENTED]	Architecture review	Client TLS authentication using per-device hardware certificates is not currently implemented in <code>ApiClient</code> .
Voice-Activated SOC Commands	[NOT IMPLEMENTED]	Architecture review	Natural language speech-to-command engine is not implemented.

3. Verified API Endpoints

The mobile client defines and interacts with the following endpoint contract (implemented in `ApiEndpoints`):

Endpoint Path	HTTP Method	Expected Request Payload / Query	Response Model / Payload	Client Handling Method
/health	GET	None	{ "status": "ok", "version": "..."} }	Environment ping & diagnostic probe
/v1/auth/login	POST	{ "email": "...", "password": "..."} }	{ "token": "...", "user": {...}} }	AuthRepository.login()
/v1/auth/me	GET	Bearer <JWT>	{ "id": "...", "name": "...", "role": "..."} }	AuthRepository.loginWithBiometrics()
/v1/dashboard/summary	GET	Bearer <JWT>	Posture score, scan counts, system health list	DashboardRepository.getDashboardSummary()
/v1/repositories	GET	Bearer <JWT>	List of monitored codebases & health scores	RepositoryRepository.getRepositories()
/v1/repositories/{id}/scan	POST	Bearer <JWT>	{ "status": "queued", "scan_id": "..."} }	RepositoryRepository.triggerRepositoryScan()
/v1/scans	GET	Bearer <JWT>	Array of historical SAST / DAST scans	ScanRepository.getScans()
/v1/findings	GET	Bearer <JWT> , optional severity query	Array of vulnerability findings & line numbers	FindingRepository.getFindings()
/v1/soc/alerts	GET	Bearer <JWT>	Array of SIEM/Wazuh alert objects	AlertsRepository.getAlerts()
/v1/soc/alerts/{id}/status	PUT	{ "status": "resolved" "investigating"} }	{ "success": true} }	AlertsRepository.updateAlertStatus()
/v1/ai/chat	POST	{ "prompt": "...", "context": "..."} }	{ "content": "...", "role": "assistant"} }	AiRepository.sendSecurityPrompt()
/v1/reports	GET	Bearer <JWT>	Array of generated compliance report metadata	ReportsScreen
/ws/alerts	WSS / WS	Query param: ? token=<JWT>	Streaming JSON alert events	WebSocketService.connect()

4. External Integrations & Integrations Architecture

4.1. Wazuh SIEM

- **Client Representation:** AlertModel with severity levels, attacker IP, destination, timestamp, and raw syslog text.

- **Mechanism:** In Demo Mode, realistic simulated Wazuh intrusion telemetry is generated locally. In Live Mode, the client connects via `/v1/soc/alerts` and `/ws/alerts` to receive forwarded Wazuh agent events.

4.2. GitHub & Semgrep SAST

- **Client Representation:** `RepositoryModel`, `FindingModel`, and `ScanModel`.
- **Mechanism:** Tracks repository branches, commit hashes, and SAST findings count. Allows triggering on-demand Semgrep scans with immediate UI acknowledgment.

4.3. Google Firebase Cloud Messaging (FCM)

- **Client Representation:** `NotificationService`.
- **Mechanism:** Registers device push tokens, handles background and foreground messages, and subscribes to enterprise threat topics (`soc_critical`, `wazuh_alerts`, `semgrep_findings`).

5. Database & Local Storage Architecture

- **Secure Credentials:** Uses `FlutterSecureStorage` to store JWT bearer tokens (`auth_token`) and custom backend URL overrides in encrypted device storage (Android Keystore / iOS Keychain).
- **Local Offline Database:** Uses `HiveStorageService` with the `securepulse_cache` box to persist recent scans, telemetry, and offline app state.
- **Backend Database:** The external FastAPI backend operates with PostgreSQL and Redis (as configured in backend deployment).

6. Authentication Pipeline

1. **Biometric Pre-Check:** Upon application launch, `BiometricService` checks hardware biometric sensor availability. If enabled, the user is authenticated via Face ID or Fingerprint.
2. **Token Check:** If authenticated, the app reads the stored JWT from `SecureStorageService` and verifies identity via `GET /v1/auth/me`.
3. **Login Form:** In case of fresh login, credentials are submitted via `POST /v1/auth/login` and returned JWT access tokens are saved to encrypted storage and assigned to `ApiClient`.
4. **Demo Mode Bypass:** When `AppConfig.isDemoMode` is enabled, an analyst profile (`Alex Vance`, Principal Security Analyst) is returned instantly without network I/O.

7. Cloud Deployment & Android Configuration

7.1. Cloud Backend

- **Primary Live Endpoint:** `https://secureguard-backend-7eqm.onrender.com` (Render Web Service).
- **Local Emulator Loopback:** `http://10.0.2.2:8000` (Android Studio emulator).
- **Localhost:** `http://127.0.0.1:8000` (Desktop / Web / iOS simulator).

7.2. Android Native Configuration

- **App ID & Label:** `com.securepulse.mobile` / `SecurePulse`.
- **Cleartext Traffic:** `android:usesCleartextTraffic="true"` configured in `AndroidManifest.xml` to allow local `http://10.0.2.2:8000` development.
- **Release Artifact:** Compiled standalone release binary `securepulse-release.apk` (63.6 MB) available in workspace root.

8. Automated Testing Validation

All automated tests have been executed via `flutter test` and verified:

```

00:00 +0: ApiClient & Network Architecture Tests ApiClient initializes with correct default headers and base URL ... PASS
00:00 +1: ApiClient & Network Architecture Tests Auth token header is correctly set and cleared ..... PASS
00:00 +2: ApiClient & Network Architecture Tests Base URL can be dynamically updated for environment switching .... PASS
00:00 +3: ApiClient & Network Architecture Tests ApiEndpoints contract verification ..... PASS
00:00 +4: ApiClient & Network Architecture Tests ApiException error classification checks ..... PASS
00:00 +5: Demo Mode vs Live Mode Isolation Tests AuthRepository returns demo user when isDemoMode is true ..... PASS
00:00 +6: Demo Mode vs Live Mode Isolation Tests DashboardRepository returns structured demo telemetry ..... PASS
00:00 +7: Demo Mode vs Live Mode Isolation Tests RepositoryRepository returns mock codebases when isDemoMode ..... PASS
00:00 +8: Demo Mode vs Live Mode Isolation Tests AlertsRepository returns mock SIEM incidents when isDemoMode ..... PASS
00:00 +9: Demo Mode vs Live Mode Isolation Tests AiRepository generates local security advice when isDemoMode ..... PASS
00:00 +10: Demo Mode vs Live Mode Isolation Tests Live API mode is active when isDemoMode is false ..... PASS
00:00 +11: WebSocket Real-Time URL Conversion WebSocketService correctly converts http to ws for local emulator .... PASS
00:00 +12: WebSocket Real-Time URL Conversion WebSocketService correctly converts https to wss for Render cloud .... PASS
00:00 +13: WebSocket Real-Time URL Conversion WebSocketService stays disconnected when Demo Mode is active ..... PASS
00:16 +14: SecurePulse Mobile app pump test ..... PASS
00:18 +15: All tests passed!

```

9. Known Limitations

1. **Firestore Configuration:** Remote push notification delivery in production requires dropping an active `google-services.json` file into `android/app/`. The app currently handles missing Firestore configurations gracefully without crashing.
2. **Live Mode Mutation Queue:** If an analyst modifies an alert status while in Live Mode without internet connectivity, the action fails with an error view rather than storing in an offline sync queue.
3. **Third-Party OAuth Redirects:** GitHub OAuth in `login_screen.dart` is currently structured as a one-tap direct auth flow rather than deep-linking through GitHub's OAuth web consent screen.